

## MATH 3310 HOMEWORK ASSIGNMENT 10

DUE ON THURSDAY 10 NOVEMBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Give a proof of the following statement:

*The inequality  $4^n > n^3$  holds for all  $n \in \mathbb{N}$ .*

- (2) Give a proof of the following statement:

*Every integer in the recursively defined sequence below is odd.*

$x_1 = 1$ ,  $x_2 = 5$ , and  $x_n = x_{n-1} + 4x_{n-2}$  for  $n \geq 3$

- (3) Recall that the Fibonacci numbers are given by

$$F_1 = 1 = F_2 \quad \text{and} \quad F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 3.$$

The number  $s = \frac{1}{2}(1 + \sqrt{5})$  is a solution to the equation  $x^2 - x - 1 = 0$ .

Use this information to prove that  $F_n \leq s^{n-1}$  holds for all  $n \in \mathbb{N}$ .

- (4) Let  $R$  be the condition on  $\mathbb{N}$  given by  $a R b$  if  $a|b$ . Determine the set

$$\{b \in \mathbb{N} \mid b R^{-1} 4\}.$$

- (5) Explain why the following argument is invalid.

*Let  $x$  and  $y$  be integers. If  $x = y$ , then one has  $xy = y^2$  and hence  $x^2 - xy = x^2 - y^2$ . This may be rewritten as*

$$x(x - y) = (x + y)(x - y),$$

*and cancellation of common factors yields  $x = x + y$ . Thus, for  $x = 1 = y$  one gets  $1 = 2$ .*